


American International University-Bangladesh (AIUB)

**Department of Computer Science**

**Faculty of Science & Technology (FST)**

# PROJECT TITLE

# BreathEase

## An AI-Powered Solution for Stress and Anxiety Management

Submitted By

| **Semester: Fall 24-25** | **Section: N** | **Group Number: 3** | |
|---|---|---|---|
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# 1. PROJECT PROPOSAL

## 1.1 Background to the Problem

In the current era, the vast advancement of technology has brought about an unprecedented change in our daily lives. In particular, the ease of access and attractive features of smartphones and social media are occupying a huge part of our time and attention. The younger generation, especially students and young professionals, spend hours after hours on platforms like Facebook, Instagram and YouTube. This excessive digital involvement is not only having a serious impact on physical health, but also on mental health.

While the benefits of technology are undeniable, its misuse has also reached alarming levels today. Studies have shown that staring at screens for a long time causes mental problems such as sleep disturbances, concentration problems, depression and anxiety (Wang et al., 2014). The results of such addiction are often not immediately understood but gradually have a negative impact on mental imbalance and emotions.

Another important aspect is that there are still a lot of prejudices and a tendency to remain silent in society about mental health. Many people do not seek treatment for symptoms of mental illness due to fear of family or society or lack of appropriate and timely advice. As a result, minor problems can turn into serious mental illnesses over time. In many cases, technology-dependent life creates such boredom that the lack of an effective personal assistant or guide to get rid of it becomes obvious

In this context, a digital and personal assistant that understands the user's behavior, helps in time management, and presents various effective ways to restore mental peace has become the need of the hour.

### Target Users and Benefits

| User Group | Key Challenges | How BreathEase Helps |
|---|---|---|
| **University Students** | Long hours on devices for study, research, leisure | Reduces stress, improves focus & sleep |
| **Young Professionals (Remote Workers)** | Long hours on devices for study, research, leisure | Prevents burnout, offers real-time calm |
| **High Social Media Users** | Digital addiction, reduced real-life interaction | Breaks doom-scrolling, builds mindfulness |
| **Individuals with Stress/Anxiety/Poor Sleep** | Emotional strain from lifestyle or overuse | Discreet, 24/7 AI-guided breathing & support |

**Key Benefit:** BreathEase empowers users by increasing self-awareness of their digital habits through contextual insights rather than just raw screen time. It provides personalized, non-intrusive nudges and recommendations to help restore balance, while offering accessible and anonymous mental wellness support free from stigma. The app equips users with practical tools for improving focus, sleep, mood regulation, and emotional resilience, enabling early intervention to prevent potential escalation of mental health issues.

## 1.2 Process Model Selection

We chose the **Scrum** Model for BreathEase app development because it is a step-by-step approach where each feature is developed in a separate Sprint. In Scrum, each Sprint has a predetermined duration (usually 1 to 4 weeks), resulting in a usable feature or part being completed within a specific time frame. This ensures clear deadlines, reduces time wastage, and makes it easier to measure progress. Daily meetings are an important aspect of Scrum, where team members provide work updates and share issues. This results in transparency within the team, faster problem resolution, and improved teamwork. As a mental health app, BreathEase has several sensitive features (such as AI suggestion, emergency alert), which require regular feedback and iteration-Scrum ensures this. Feature-based division, time-bound Sprints, and team-based communication make this model the most suitable for BreathEase.

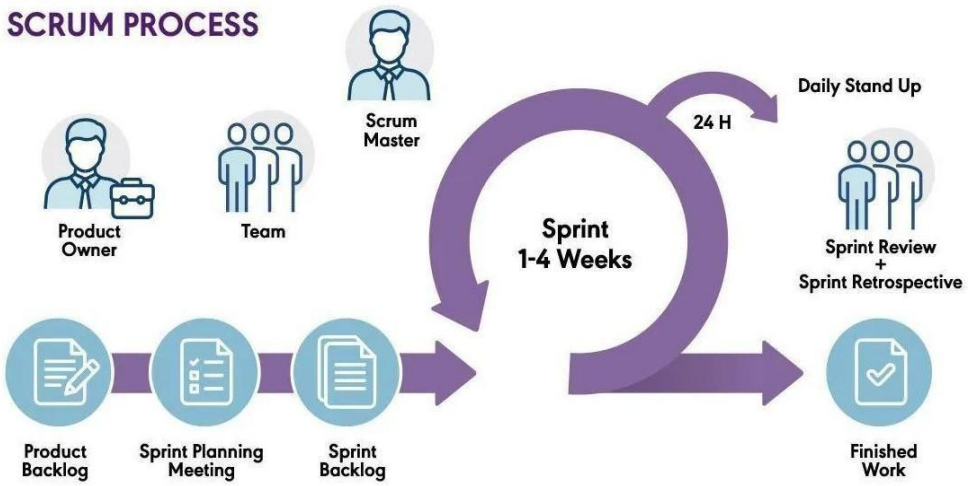



Image Source: https://www.pm-partners.com.au/insights/the-agile-journey-a-scrum-overview/

**Why Scrum Model Was Chosen for BreathEase**

The Scrum Model was selected for the BreathEase application because it effectively supports the development of a feature-rich and sensitive mental health system. BreathEase includes critical functionalities such as AI-based mental health suggestions, emergency alert mechanisms, and continuous user monitoring, which require careful development and frequent validation. Scrum allows these features to be developed and tested individually within short, time-boxed Sprints, ensuring better focus and higher quality.

Another key reason for choosing Scrum is its flexibility. Mental health applications must continuously evolve based on user feedback, expert recommendations, and real-world usage. Scrum supports requirement changes even in later stages of development without disrupting the overall workflow. The model also emphasizes regular feedback and iteration, enabling the team to review completed features at the end of each Sprint and improve them continuously.

Scrum's time-bound delivery structure helps maintain clear schedules and measurable progress, reducing delays and improving productivity. Moreover, daily Scrum meetings promote transparency, strong collaboration, and quick problem resolution among team members. Finally, by delivering working features early and regularly, Scrum helps identify and reduce risks related to data privacy, system reliability, and emergency response, making it a highly suitable process model for the BreathEase app.

**Why Not other process Model?**

There are several software development process models, but not all are suitable for every project. The Waterfall model follows a rigid sequential approach and works well only when requirements are fixed, making it unsuitable for BreathEase where frequent changes and user feedback are essential. The V-Model improves quality through parallel testing but still lacks flexibility and makes rapid feature updates difficult. The Incremental model offers partial flexibility but does not support continuous team communication or timely feedback integration. The Prototype model is effective for small systems but can waste time and resources on large, complex projects. Extreme Programming (XP) works best for small teams with limited scope and is not ideal for a feature-rich application like BreathEase. DSDM ensures fast delivery but is complex and better suited for large organizations, making it less practical for a mid-sized project like BreathEase.

## Visualization of Scrum Model

• **Product Backlog** – A prioritized list of all features and requirements, maintained by the Product Owner. *Example: User registration, breathing exercises, AI suggestions, emergency alerts.*

• **Sprint Planning** – The team selects backlog items to complete in one Sprint, forming the Sprint Backlog.

• **Sprint (Development Phase)** – Design, coding, testing, and integration are performed. Each Sprint delivers a potentially shippable product increment**.**

• **Daily Scrum** – A short daily meeting to discuss progress, plans, and obstacles, improving communication and transparency**.**

• **Sprint Review** – Completed features are demonstrated to stakeholders, and feedback is added to the Product Backlog.

• **Sprint Retrospective** – The team evaluates what went well, what needs improvement, and plans how to enhance the next Sprint.

## 2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

### 2.1 SCOPES & FEATURES

**Behavior Monitoring** – BreathEase tracks your digital habits, like app usage and late-night activity, to identify patterns affecting your mental well-being.

**Smart Alerts & Pattern Detection** – The app sends gentle reminders when usage is excessive or at unhealthy hours, encouraging breaks and better sleep.

**Mental Wellness Toolkit** – Includes guided yoga and breathing exercises, mood tracking, digital journaling, soothing music, and short skill-building lessons to support calmness and focus.

**Anonymous Consultation** – Connect with licensed therapists via private chat or voice calls without revealing your identity.

**AI-Based Personal Assistant** – Provides reminders, tracks habits and goals, and offers motivational feedback to support healthy routines.

**Emergency Support & Crisis Detection** – Detects distress signals and guides users to helplines, SOS contacts, or immediate expert help.

**Personalized Goals & Progress Tracker** – Set small goals, track progress visually, and receive encouragement to build healthier habits over time.

## 2.2 User Story Table

| As a | I want to | So that | Acceptance Criteria |
|---|---|---|---|
| **1.University Student** | Understand and manage my daily screen usage | I can maintain focus, reduce stress, and avoid unhealthy habits | □ System tracks contextual screen usage (study/social/entertainment)<br>□Provides personalized insights instead of raw screen time numbers<br>□ Sends non-intrusive reminders during long usage periods<br>□ Offers focus, breathing, and stress-relief tools |
| **2.Young Professional / Remote Worker** | Balance my screen-heavy work routine | I can prevent burnout and improve emotional wellbeing | □ Detects long work sessions and suggests breaks<br>□ Provides stress-relief and mindfulness exercises<br>□ Gives daily wellness summaries<br>□ Includes focus-mode / quiet-mode features |
| **3.High Social Media User** | Control my social media scrolling habits | I can reduce addiction and build healthier routines | □ Identifies doom-scrolling patterns<br>□ Suggests short mindfulness or breathing activities<br>□ Tracks progress to show improvement |
| **4.Individual with Stress/Anxiety/Sleep Issues** | Access-guided relaxation & emotional support | I can improve mood, sleep, and mental resilience | □ Provides 24/7 breathing, calming, and relaxation sessions<br>□ Includes mood tracking and daily check-in<br>□ Suggests personalized routines for sleep and stress |

## 2.3 Requirements Traceability Matrix

### Functional Requirements

| ID No | Requirement Description | Source | Design | Verification Method | Owner |
|---|---|---|---|---|---|
| FR-01 | User can sign up using email or social accounts | SRS | Use Case Diagram, UI Page 1 & 4 | Test Case (TC-01, TC-04) | Developer |
| FR-02 | Registered user can log in using valid credentials | SRS | Use Case, Sequence Diagram | Test Case (TC-02) | Developer |
| FR-03 | User can continue without sign up with limited access | SRS | Use Case Diagram, UI Page 1 | Functional Testing | Product Owner |
| FR-04 | User can recover password using email | SRS | Sequence Diagram | Test Case (TC-02) | Developer |
| FR-05 | App requests permission for behavior monitoring | SRS | Activity Diagram | Test Case (TC-07) | Developer |
| FR-06 | App tracks screen time and app usage | SRS | Class Diagram, Dashboard UI | Test Case (TC-07) | Developer |
| FR-07 | Smart alerts generated for unhealthy usage | SRS | Activity Diagram | Test Case (TC-06) | AI Module Team |
| FR-08 | Mental wellness toolkit (yoga, breathing, music) available | SRS | Use Case Diagram, UI Page 6 | Test Case (TC-06) | Developer |
| FR-09 | User can set and manage personal goals | SRS | UI Page 3, Class Diagram | Test Case (TC-03) | Developer |
| FR-10 | AI provides personalized suggestions | SRS | Class Diagram (AI Engine) | Functional Testing | AI Module Team |
| FR-11 | Anonymous consultation with therapist | SRS | Use Case, Sequence Diagram | Test Case (TC-08) | Therapist Module Team |
| FR-12 | Emergency support triggered during crisis | SRS | Activity Diagram | Scenario Testing | AI Module Team |

| | | | | | |
|---|---|---|---|---|---|
| FR-13 | User profile can be viewed and updated | SRS | UI Page 5 | Test Case (TC-05) | Developer |
| FR-14 | Dashboard displays all major features | SRS | Dashboard UI Design | Test Case (TC-06) | Developer |
| FR-15 | Users can submit feedback or report issues | SRS | Use Case Diagram | Manual Testing | Product Owner |

**Non-functional Requirements**

| ID No | Requirement Description | Source | Design | Verification Method | Owner |
|---|---|---|---|---|---|
| NFR-01 | System must ensure user data privacy and confidentiality | SRS | System Architecture, Security Design | Security-Testing, Encryption Validation | Security Team |
| NFR-02 | System must provide secure data storage and transmission | SRS | Database Design,API Security | Penetration Testing | Security Team |
| NFR-03 | Application should respond within acceptable time | SRS | Performance Architecture | Performance Testing | Developer |
| NFR-04 | System should handle multiple users simultaneously | SRS | Cloud& Scalability Design | Load Testing | DevOps Team |
| NFR-05 | Application must be highly available | SRS | Cloud Deployment Design | Availability Monitoring | DevOps Team |
| NFR-06 | User interface must be easy and user-friendly | SRS | UI/Wireframe Design | Usability Testing | UI/UX Team |
| NFR-07 | Applications should work smoothly on different devices | SRS | Cross-Platform Design | Compatibility Testing | QA Team |
| NFR-08 | System should be reliable with minimal crashes | SRS | Fault-Tolerance Design | Stress Testing | QA Team |
| NFR-09 | Applications should support future feature expansion | SRS | Modular Architecture | Review & Inspection | Architect |
| NFR-10 | System must protect against unauthorized access | SRS | Authentication & Authorization Design | Security Testing | Security Team |

### 2.3.1 Functional Requirements

1. **Software Sign up**

Functional Requirements:

- New users will get a signup page.
- Signup page will have username, email, and password text fields.
- The system will validate email and username.
- If email/username already exists, a popup will ask the user to enter different credentials.
- After successful signup, user details will be securely stored in the database.

Priority Level: Medium

Precondition: Email and username must be unique.

2. **Software Log in**

Functional Requirements:

- Registered users can Log in via username/email and password.
- The app will check credentials from the database.
- After successful login, user is redirected to the main dashboard.

Priority Level: High

Precondition: User must have a registered account.

3. **Behavior Monitoring Access**

Functional Requirements:

- Apps will ask permission to monitor digital activities (screen time, app usage).
- A popup prompt will request user consent for tracking.
- Users can allow or deny permission.
- Without permission, monitoring features will remain disabled.

Priority Level: High

Precondition: Device must allow activity tracking.

4. **Continue without Sign in/Sign up**

Functional Requirements:

- Users can enter the app without registration.
- They will have limited access to wellness resources (e.g., breathing exercises, music).

Priority Level: Medium

Precondition: Limited features only; no personalized tracking.

5. **Forgotten Password**

Functional Requirements:

- "Forgot Password" option on login page.
- User enters registered email.
- System sends a password reset link.

Priority Level: Medium

Precondition: User must have a registered email.

6. **Reset Password**

Functional Requirements:

- Page to reset password after clicking email link.
- Users must enter new passwords and confirm it.
- System validates and updates databases.

Priority Level: Medium

Precondition: User must access reset link from email.

7. **Digital Behavior Tracking**

Functional Requirements:

- Tracks phone usage duration, app usage, and late-night activity.
- Data will be displayed on the dashboard.
- Updates periodically in real-time.

Priority Level: High

Precondition: User must grant monitoring permission.

8. **Smart Alerts and Notifications**

Functional Requirements:

- Sends reminders for excessive usage or unhealthy timing.
- Example: "You've been scrolling for 2 hours, take a break."
  Notifications include text and friendly suggestions.

Priority Level: High

Precondition: Notification access required.

9. **Mental Wellness Toolkit**

Functional Requirements:

- Provides guided yoga and breathing exercises.
- Includes mood tracker, journaling, and relaxing sounds.
- Available both online and offline.

Priority Level: High

Precondition: None

10. **Anonymous Consultation Services**

Functional Requirements:

- Users can chat/call with licensed therapists.
- Communication will be private and anonymous.
- Data security ensured via encryption.

Priority Level: High

Precondition: Internet connection required.

11. **AI-Based Personal Assistant**

Functional Requirements:

- Apps give personalized advice based on user behavior.
- Tracks daily habits and provides motivational feedback.
- Suggest healthy routines like sleep, yoga, and breaks.

Priority Level: High

Precondition: User must grant monitoring access.

12. **Emergency Support & Crisis Detection**

Functional Requirements:

- Detects distress from mood tracker/journal entries.
- Sends alerts with helpline numbers or SOS contacts.

  Provides emergency resources immediately.

Priority Level: High

Precondition: None

13. **Personalized Goals & Progress Tracker**

Functional Requirements:

- Users can set personal goals (reduce screen time, meditation).
- App tracks progress with visual reports.
- Sends motivational reminders.

Priority-Level:Medium

Precondition: User Logged in

14. **Community Feedback & Reports**

Functional Requirements:

- Users can share feedback or experience.
- Option to report harmful digital behavior anonymously.
- Data used to improve recommendations.

Priority Level: Medium

Precondition: Internet required.

### 2.3.2 Non-Functional Requirements

**Feasibility:**

This app is very easy to create with today's technology. Using the AI capabilities of modern mobile phones, cloud databases such as Firebase or MongoDB, and tools like Flutter or React Native, the app can be developed quickly and effectively. As a result, users will get accurate and smooth experience.

**Innovation:**

Many apps can measure screen time, but the specialty of BreathEase is that it understands the user's behavior and extends a helping hand accordingly. In addition, it allows you to talk to therapists anonymously, which is a great help in mental health for many.

**Impact Potential:**

Using this app will have good mental health, increase work efficiency, sleep better, and improve quality of life. In the long run, it will help improve our quality of life.

**User Acceptance:**

Users will easily like this app and use it regularly due to its simple and friendly design and privacy protection.

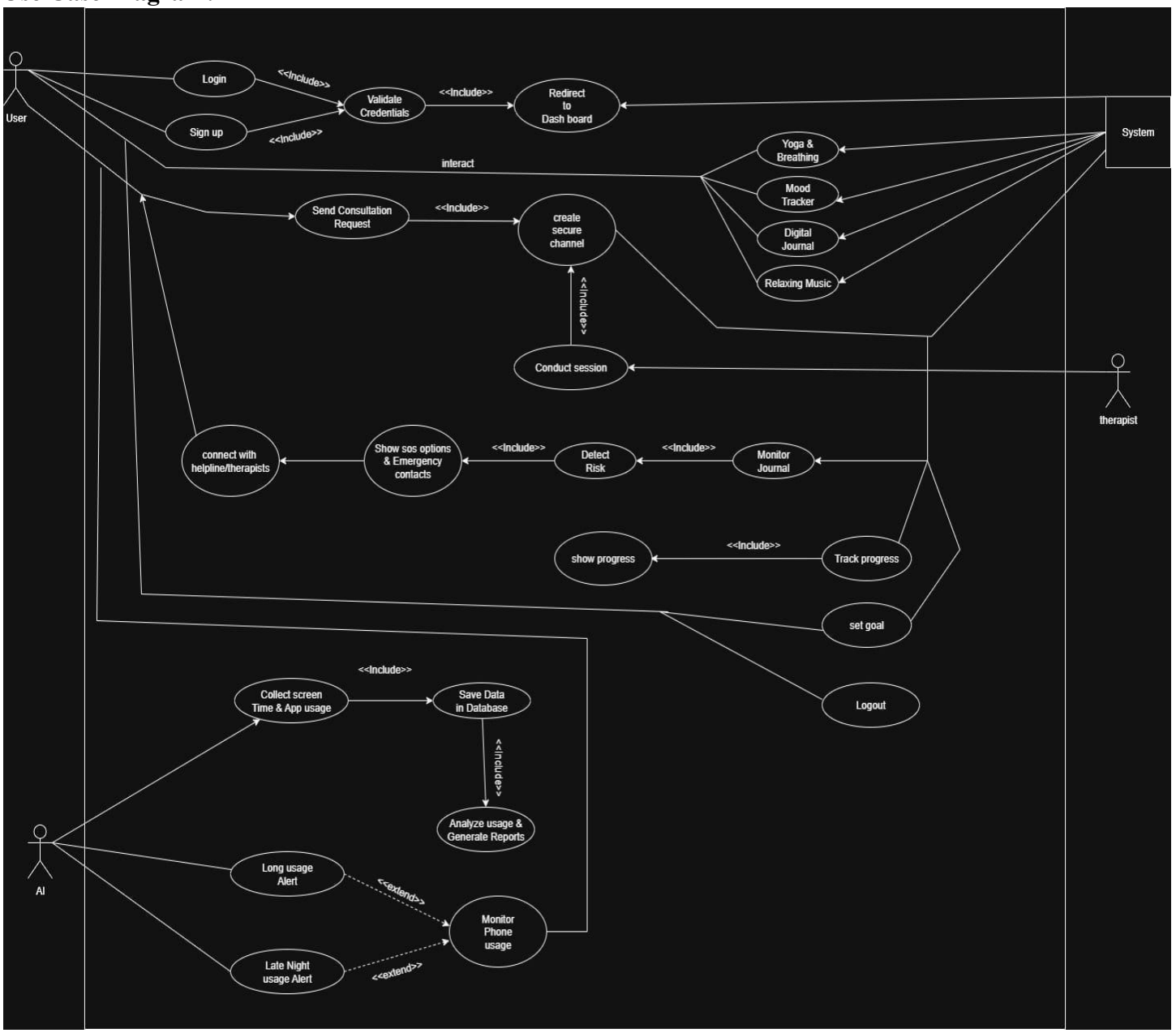
**Cost-Effectiveness:**

This app can be created using advanced technology at low cost, so many people will be able to easily get mental health support

# 3. SOFTWARE DESIGN

## 3.1 System Design

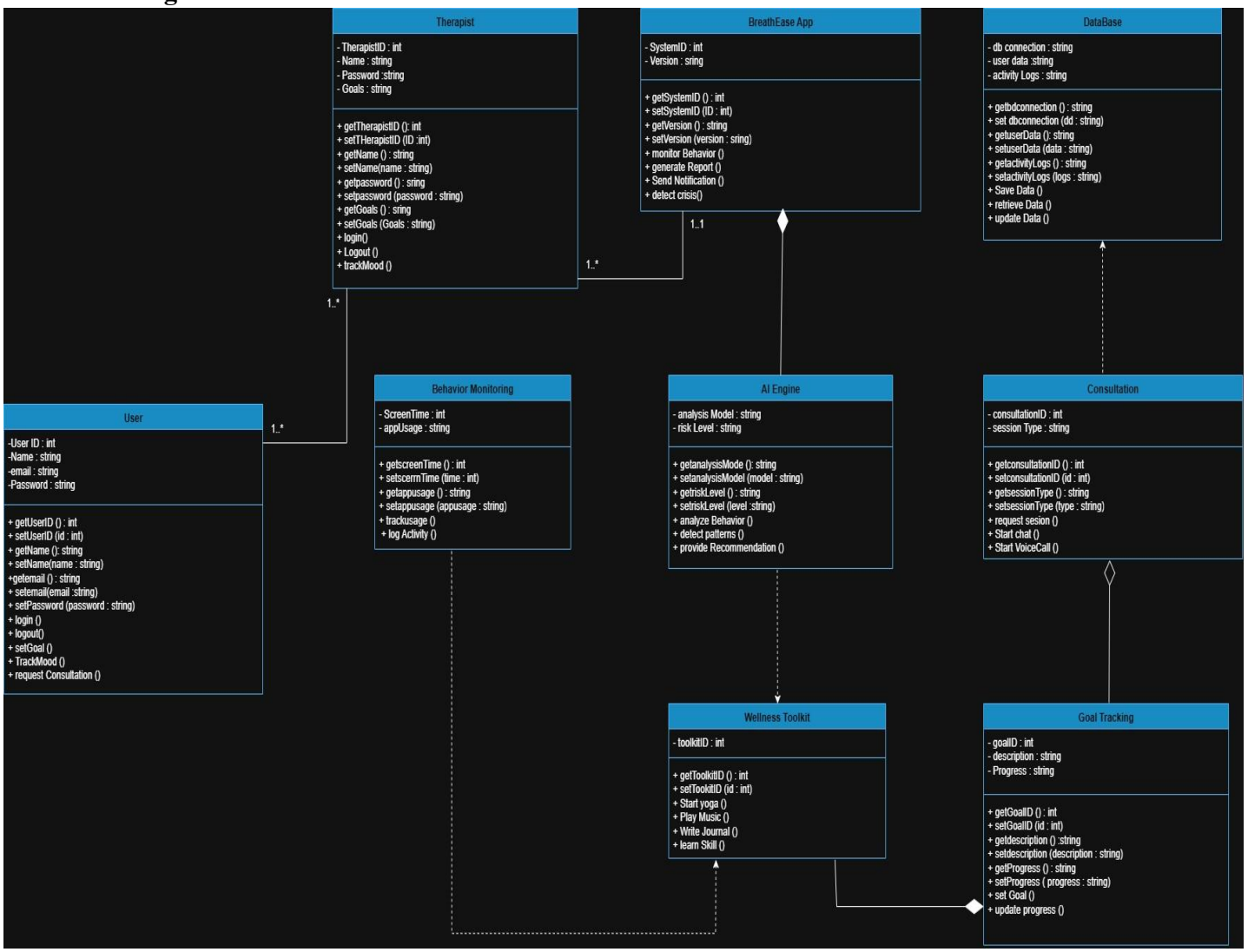
**Use Case Diagram:**




## Description:

The diagram outlines the user interactions and system functions for a "BreathEase App". The main actors are the User and the Therapist. A User can register, manage their profile, track their mood, and book consultations. The app's core functionality, powered by an AI Engine, allows the system to monitor behavior, determine risk, and provide personal recommendations to the user, who can also access a Wellness Toolkit. The Therapist can manage their own profile and view the user's data to provide support

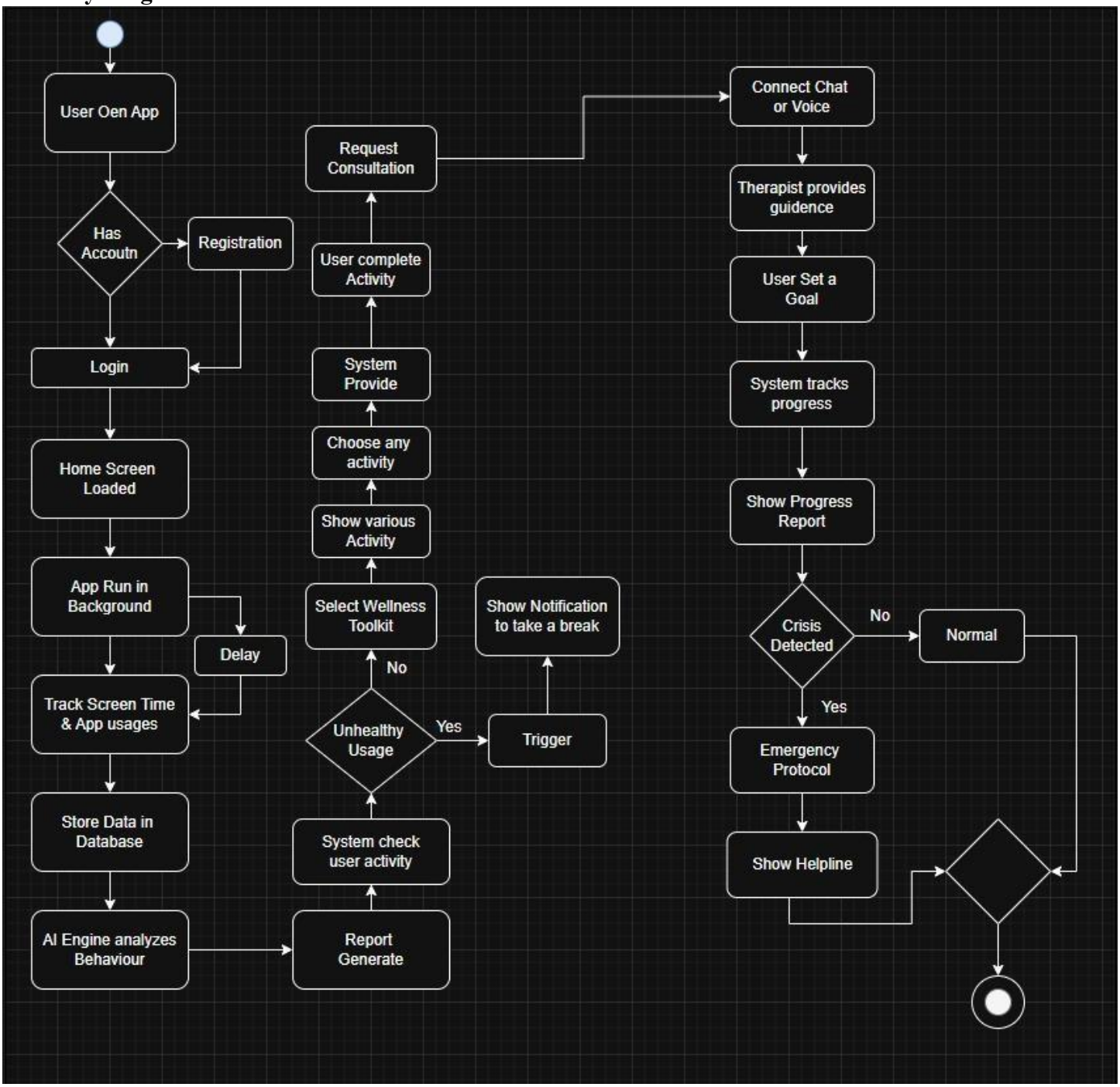
## ✞ Class Diagram:




## Description:

The diagram shows relationships like association (e.g., User to Behavior Monitoring, User to Wellness Toolkit), aggregation (e.g., Therapist to User), and dependency (e.g., AI Engine to Behavior Monitoring). The numbers (e.g., "1..*", "1.1") indicate multiplicity, showing how many instances of one class can be related to instances of another. For instance, one User can be associated with multiple Behavior Monitoring records.

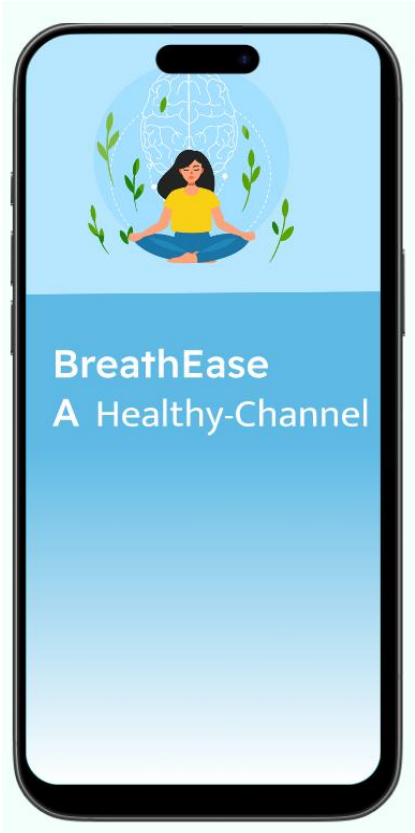
**Activity Diagram:**



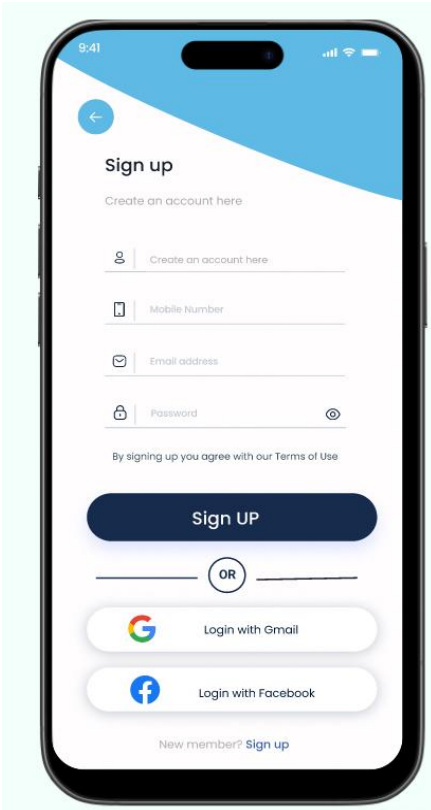

**Description:** The activity diagram shows how the BreathEase App supports users through AI monitoring and therapist consultation. After registration or login, the system tracks screen time, stores data, and analyzes behavior. If unhealthy usage is detected, notifications and wellness activities are provided. Users can request consultation, set goals with therapist guidance, and track progress. In case of crisis, the system triggers emergency protocols and displays the helpline; otherwise, users continue normally.

## 4.2 UI / Wireframe Design UI

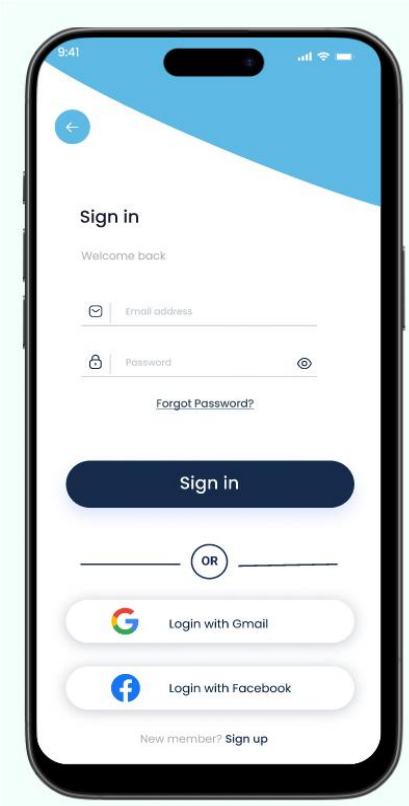
## Design (Figma):



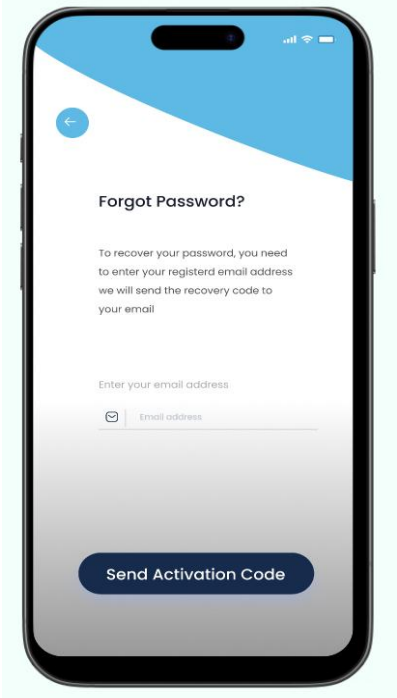

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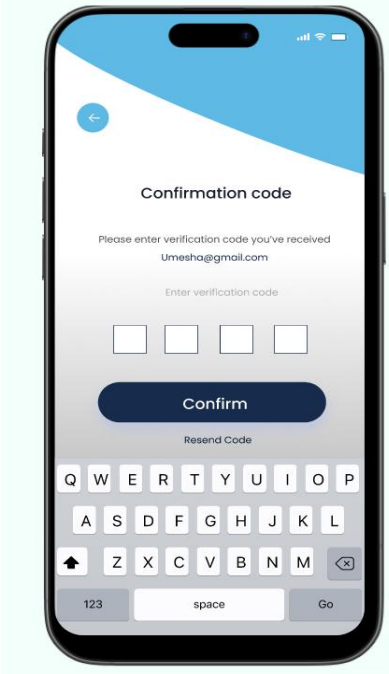

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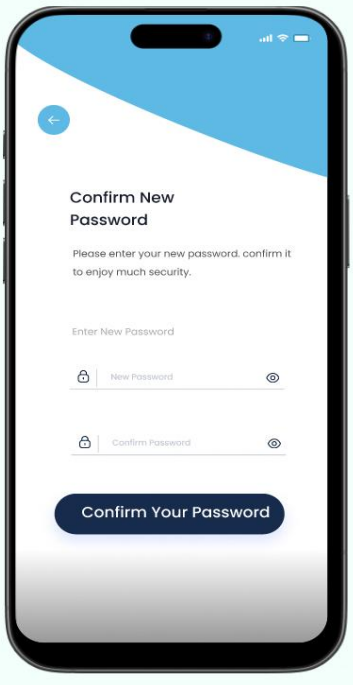

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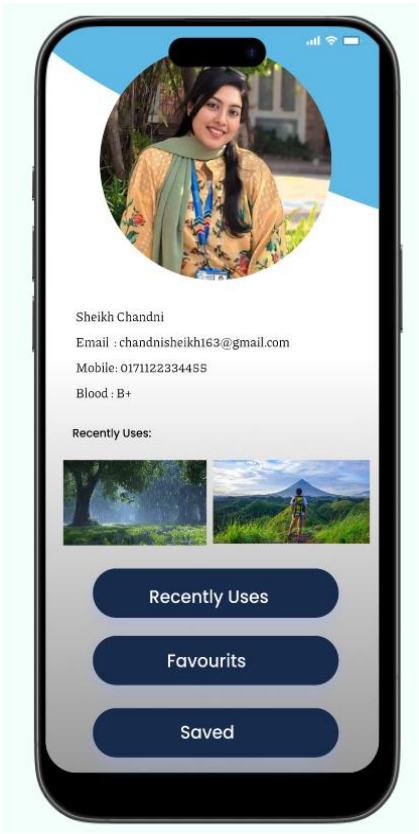

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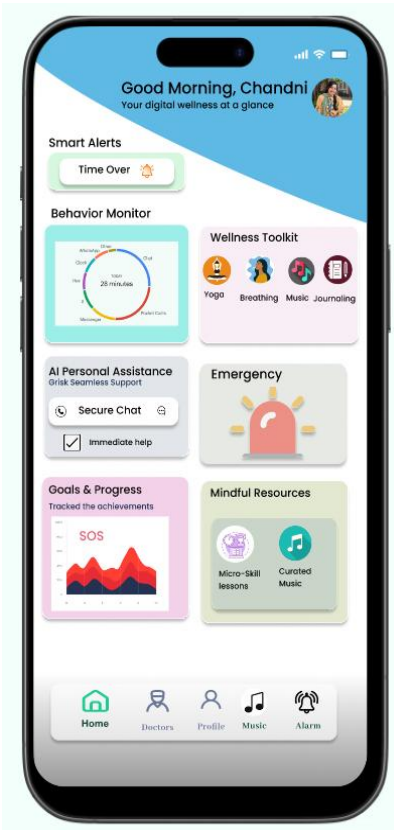

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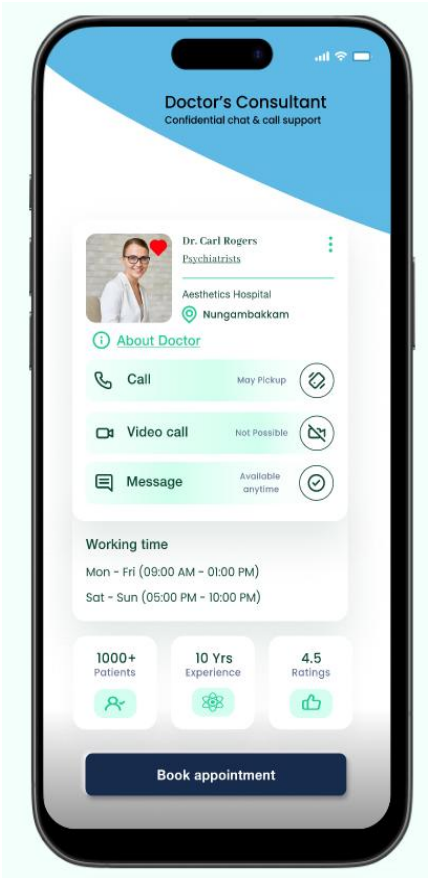

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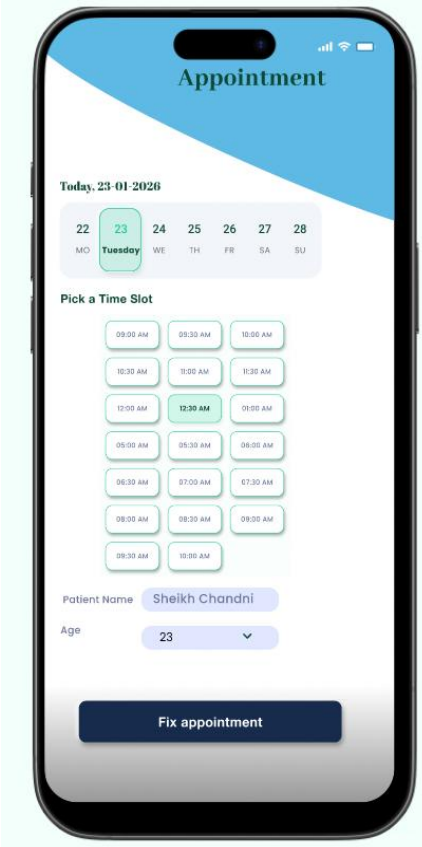

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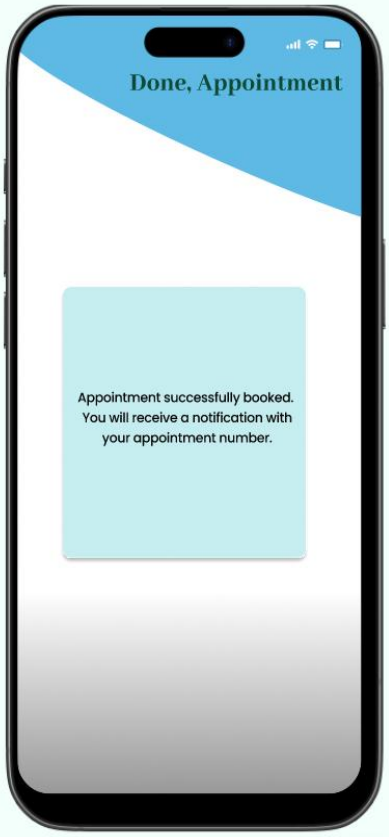

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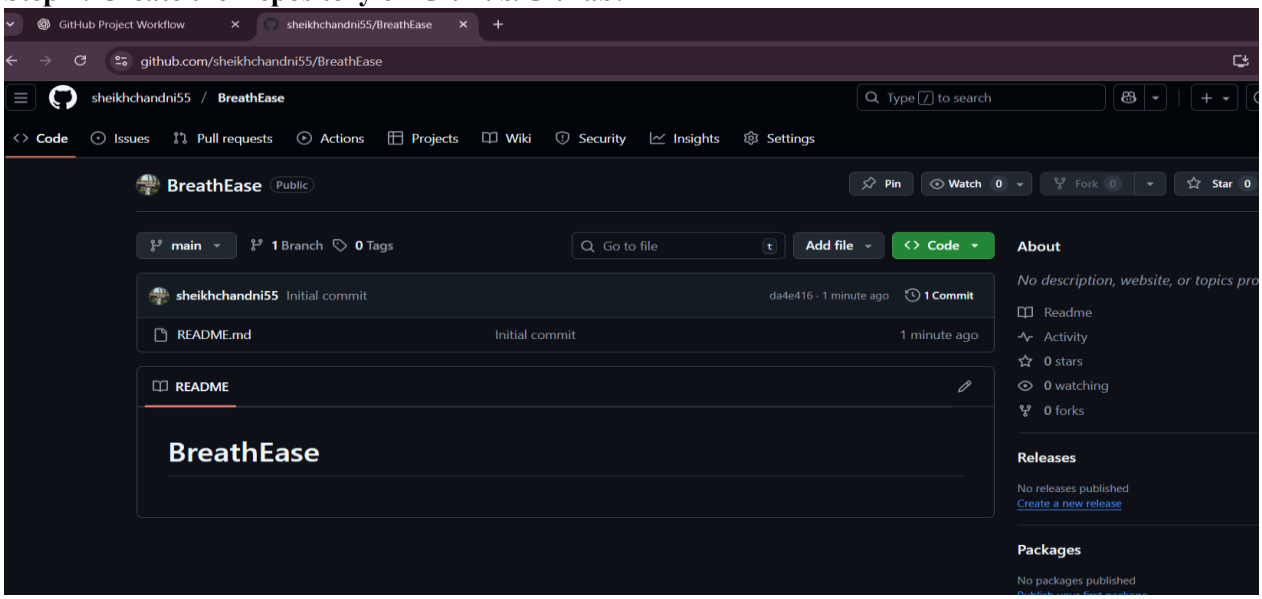

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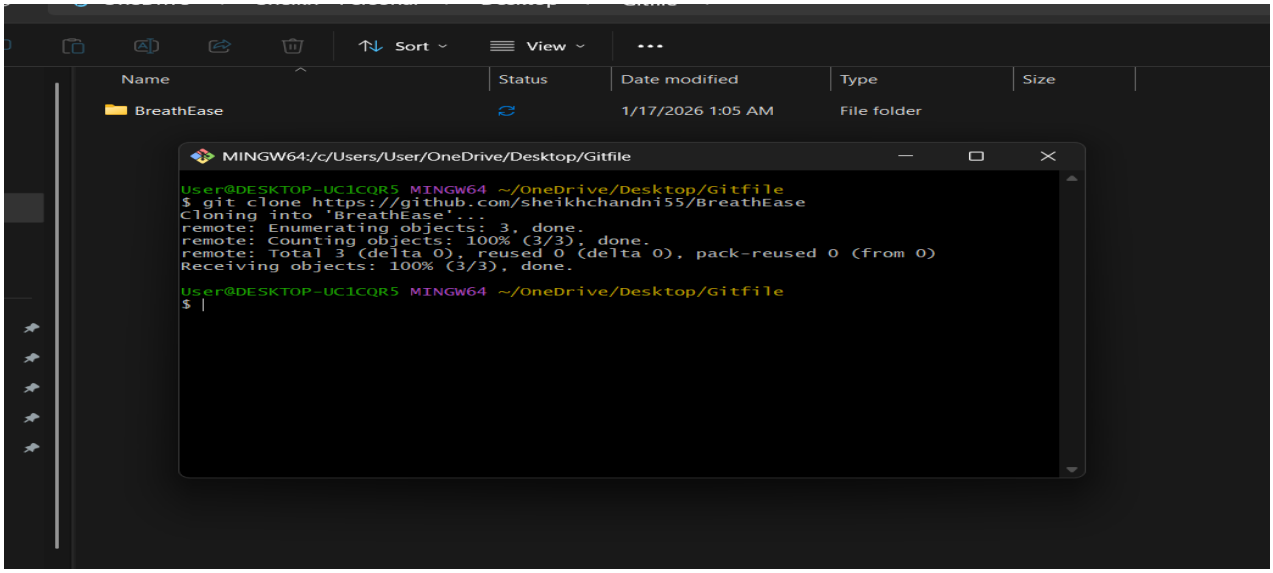

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## 5. GIT WORKFLOW

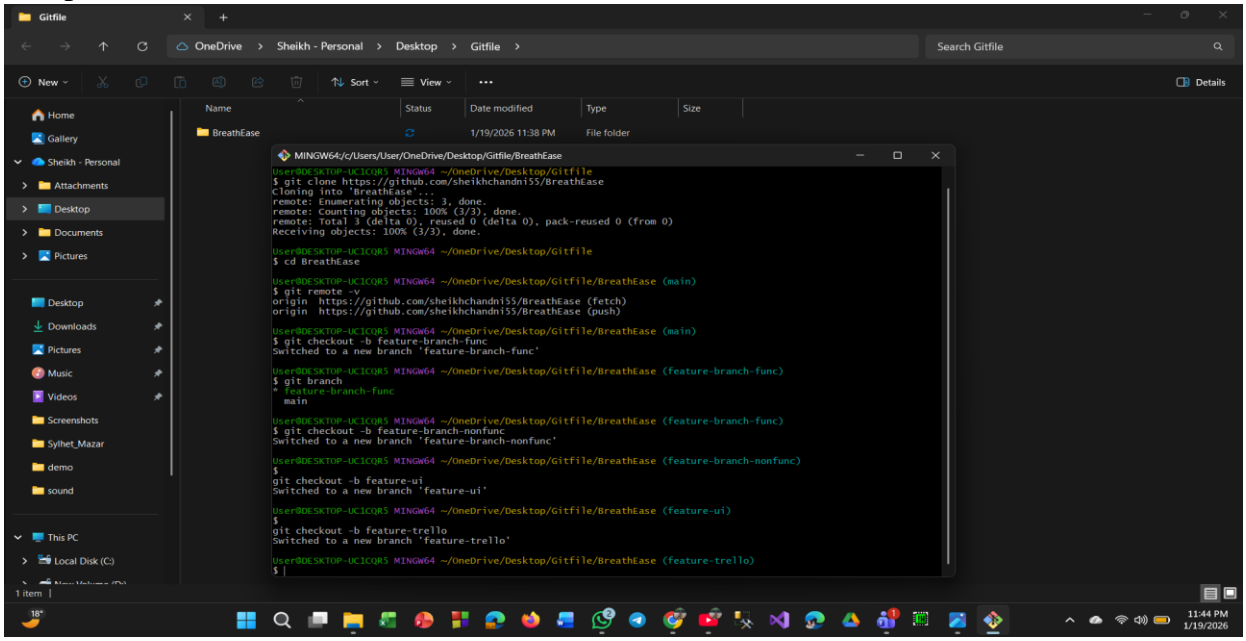
### Step 1: Create the Repository on GitHub/GitLab:



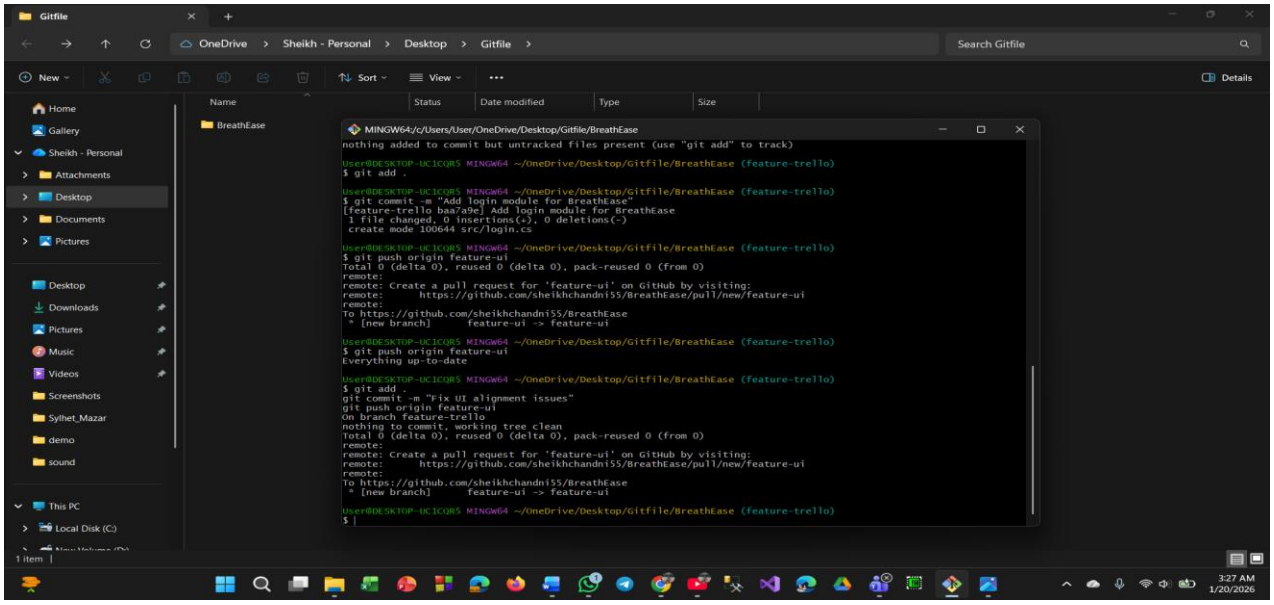

### Step 2: Initialize the clone Repository Locally and Create the Structure:



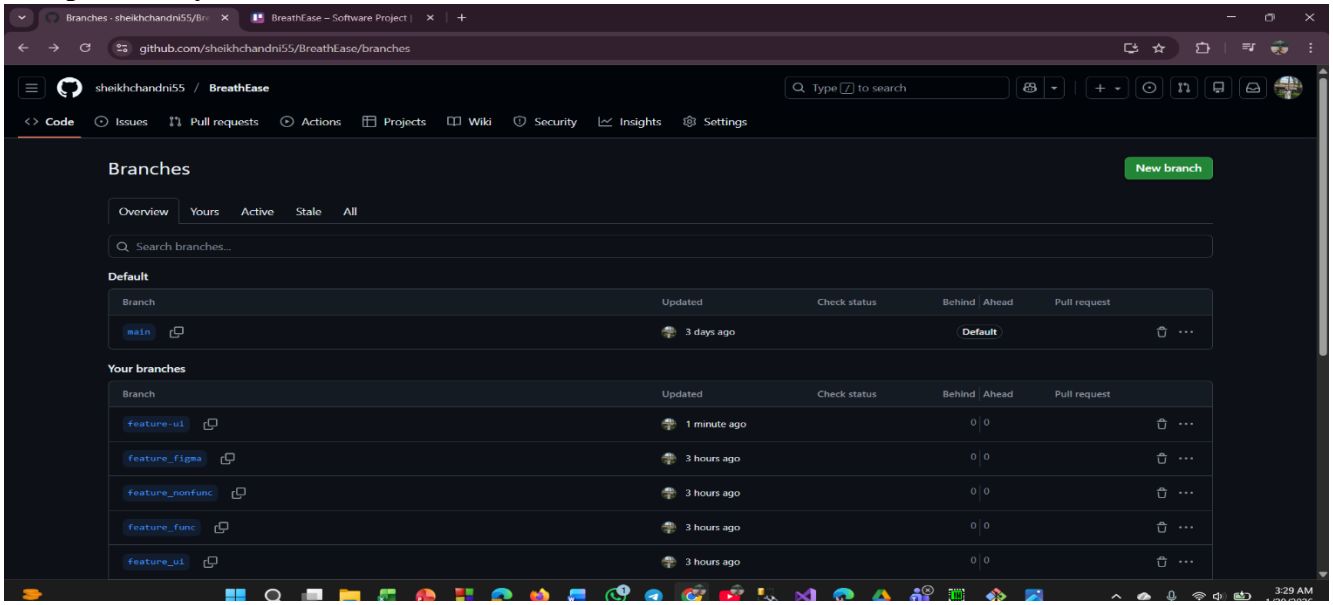
## Step 3: Rename the Default Branch and Create the Workflow Branches:



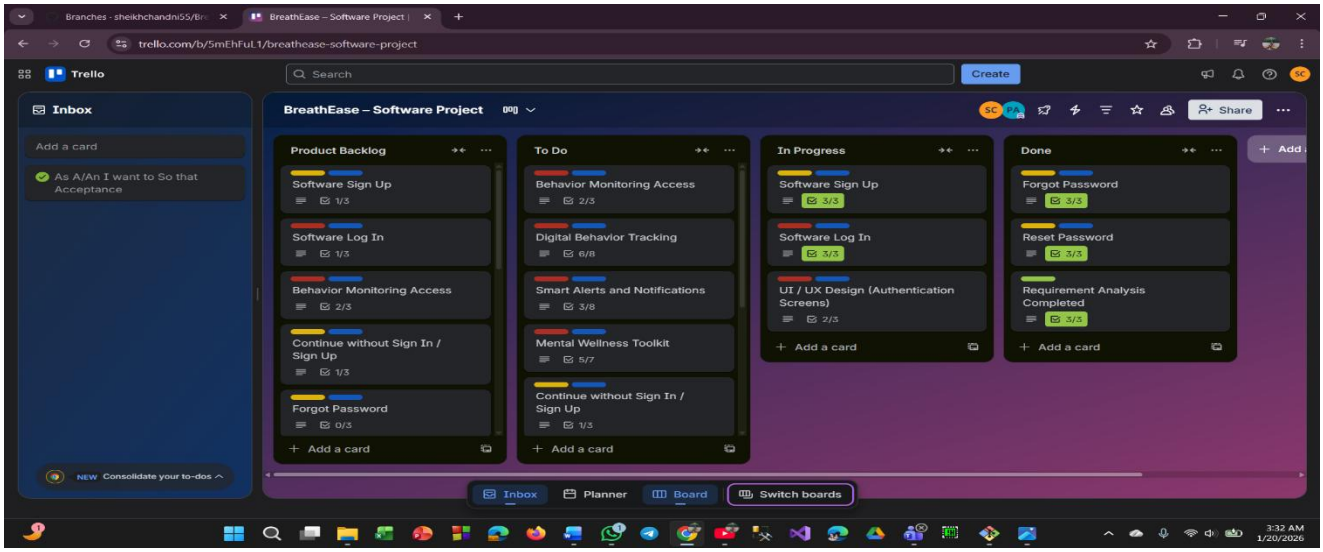

## Step 4: Connect the Local Repository to the Remote and Push All Branches:



## Step 5: Verify Success:




## Trello workflow:




## Figma workflow:

https://www.figma.com/design/XYsnPWf8X0vQMYZuyEaRng/BreathEase?node-id=4-370&t=5lSYzMjavVWkCIR2-0

## 6. TESTING

| **Project Name:** BreathEase | | | **Test Designed by: Sheikh Chandni** | |
|---|---|---|---|---|
| **Test Case ID:** TC_01 | | | Test Designed date: 15/01/2026 | |
| **Test Priority (Low, Medium, High): Medium** | | | Test Executed by: Jahed Rakib | |
| **Module Name:** Welcome / Sign Up Options | | | Test Execution date: 16/01/2026 | |
| **Test Title:** Verify sign-up using Gmail, Facebook | | | | |
| **Description:** Ensure users can sign up through different available options. | | | | |
| **Precondition:** App installed, internet available. | | | | |
| **Dependencies:** OAuth services (Gmail, Facebook). | | | | |
| **Test Steps** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |
| 1. Open BreathEase app.<br>2. Click “Sign Up with Gmail”<br>3. Click “Sign Up with Facebook”. | 1. Email: test@gmail.com Password: 12345<br>2.Valid Gmail account<br>3. Valid Facebook account | 1.Welcome page loads<br>2.Redirects to Create Account<br>3.Logs in successfully | As expected | Pass |

| **Project Name:** BreathEase | **Test Designed by: Prantik Paul** |
|---|---|
| **Test Case ID:** TC_02 | Test Designed date: 16/01/2026 |
| **Test Priority (Low, Medium, High): High** | Test Executed by: Abul Kashem Md Rahat |
| **Module Name:** Sign up | Test Execution date: 19/01/2026 |
| **Test Title:** Verify Sign up with valid and invalid credentials. | |
| **Description:** Verify login functionality | |

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|---|---|---|---|---|
| **Precondition:** User has a create account | | | | |
| **Dependencies:** Database availability. | | | | |
| **Test Steps** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |
| 1. Enter valid email, password ,mobile number<br>2. Enter wrong password<br>3. Leave fields blank | 1 .Email: test@mail.com Password: 12345<br>2. Email: test@mail.com, Pass: wrong | 1.Logs in successfully<br>2. Error message<br>3. Validation error shown | As expected | Pass |

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|---|---|---|---|---|
| **Project Name:** BreathEase | | | **Test Designed by: Sheikh Chandni** | |
| **Test Case ID:** TC_03 | | | Test Designed date: 18/01/2026 | |
| **Test Priority (Low, Medium, High): High** | | | Test Executed by: Jahed Rakib | |
| **Module Name:** Sign in | | | Test Execution date: 19/01/2026 | |
| **Test Title:** Verify account creation | | | | |
| **Description:** Verify account creation with valid details. | | | | |
| **Precondition:** User not yet registered. | | | | |
| **Dependencies:** Email must be unique. | | | | |
| **Test Steps** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |

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|---|---|---|---|---|
| 1. Enter Email, Password, Confirm Password<br>2. Click Submit<br>3. Enter existing email | 1.Name: Alex, Email: alex@test.com, Pass: 12345<br>2.Valid data<br>3.Email already used | 1.Data accepted<br><br>2.Account created and stored in DB<br>3.Error message displayed | As expected, | Pass |

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|---|---|
| **Project Name:** BreathEase | **Test Designed by: Rahat** |
| **Test Case ID:** TC_04 | Test Designed date: 11/01/2026 |
| **Test Priority (Low, Medium, High): High** | Test Executed by: Rakib |
| **Module Name:** Forgot Password | Test Execution date: 15/01/2026 |
| **Test Title:** Verify password | |
| **Description:** Verify that a user can reset their password | |
| **Precondition:** User must have an existing account registered with a valid email. | |
| **Dependencies:** User email must be registered in the system. | |

| **Test Step** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |
|---|---|---|---|---|
| 1. Click on “Forgot Password” link<br>2. Enter registered email and submit | 1. Email: alex@test.com | 1. Password reset link sent to registered email | As expected, | Pass |

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|---|---|
| **Project Name:** BreathEase | **Test Designed by: Prantik Paul** |
| **Test Case ID:** TC_05 | Test Designed date: 14/01/2026 |
| **Test Priority (Low, Medium, High): High** | Test Executed by: Sheikh Chandni |

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|---|---|
| **Module Name:** Password Reset | Test Execution date: 18/01/2026 |
| **Test Title:** Verify Confirmation Code (OTP) Functionality | |
| **Description:** Verify that the system correctly handles confirmation codes (OTP) | |
| **Precondition:** Email or phone service is working to deliver the code. | |
| **Dependencies:** OTP has a valid expiry time | |

| Test Steps | Test Data | Expected Results | Actual Results | Status (Pass/Fail) |
|---|---|---|---|---|
| 1. Enter correct OTP received via email<br>2. Enter expired OTP<br>3. Request a new OTP | 1. OTP: 123456<br>2. OTP received > 1 min ago<br>3. N/A | 1. OTP verified successfully; user proceeds to next step<br>2. Error message displayed: "Confirmation code expired" | As expected | Pass |

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| **Project Name:** BreathEase | **Test Designed by: Sheikh Chandni** |
| **Test Case ID:** TC_06 | Test Designed date: 18/01/2026 |
| **Test Priority (Low, Medium, High): High** | Test Executed by: Jahed Rakib |
| **Module Name:** Password Reset | Test Execution date: 19/01/2026 |
| **Test Title:** Verify Confirmation of New Password | |
| **Description:** Verify that a user can successfully set a new password | |
| **Precondition:** User not yet registered. | |
| **Dependencies:** Email must be unique. | |

| Test Steps | Test Data | Expected Results | Actual Results | Status (Pass/Fail) |
|---|---|---|---|---|
| 1.Enter Email, Password, Confirm Password<br>2.Click Submit<br>3.Enter existing email | 1.Name: Alex, Email: alex@test.com, Pass: 12345<br>2.Valid data<br>3.Email already used | 1.Data accepted<br>2.Account created and stored in DB<br>3.Error message displayed | As expected, | Pass |

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|---|---|---|---|---|
| **Project Name:** BreathEase | | | | **Test Designed by: Sheikh Chandni** |
| **Test Case ID:** TC_07 | | | | Test Designed date: 18/01/2026 |
| **Test Priority (Low, Medium, High): High** | | | | Test Executed by: Rahat |
| **Module Name:** Profile | | | | Test Execution date: 19/01/2026 |
| **Test Title:** Verify profile view and update | | | | |
| **Description:** Ensure users can view and update their profile. | | | | |
| **Precondition:** User logged in. | | | | |
| **Dependencies:** Database must store changes | | | | |
| **Test Steps** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |

| | | | | |
|---|---|---|---|---|
| 1. Open Profile page<br>2. Edit information<br>3. Save changes | 1. Update Name, Favorites 2. Save button | 1. Profile details shown<br>2. Info editable<br>3. Profile updated successfully | As expected, | Pass |

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|---|---|---|---|---|
| **Project Name:** BreathEase | | | | **Test Designed by: Jahed Rakib** |
| **Test Case ID:** TC_08 | | | | Test Designed date: 15/01/2026 |
| **Test Priority (Low, Medium, High): High** | | | | Test Executed by: Sheikh Chandni |
| **Module Name:** Dashboard | | | | Test Execution date: 16/01/2026 |
| **Test Title:** Verify dashboard loading and features | | | | |
| **Description:** Ensure dashboard loads and shows all modules. | | | | |
| **Precondition:** User logged in. | | | | |
| **Dependencies:** All modules integrated | | | | |
| **Test Steps** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |
| 1.Login and navigate to Dashboard<br>2.Check Behavior Monitor section<br>3.Check Smart Alerts<br>4.Check Wellness Toolkit | | 1. Dashboard loads<br>2. Shows screen time & app usage<br>3. Alerts displayed<br>4. Tools accessible | As expected | Pass |

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| **Project Name:** BreathEase | **Test Designed by: Rahat** |
| **Test Case ID:** TC_09 | Test Designed date: 12/01/2026 |

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|---|---|
| **Test Priority (Low, Medium, High): High** | Test Executed by: Prantik Paul |
| **Module Name:** Behavior Monitor | Test Execution date: 13/01/2026 |
| **Test Title:** Verify behavior monitoring | |
| **Description:** Ensure screen time and app usage are tracked correctly | |
| **Precondition:** User granted permission | |
| **Dependencies:** Device activity tracking enabled | |

| Test Step | Test Data | Expected Results | Actual Results | Status (Pass/Fail) |
|---|---|---|---|---|
| 1. Grant tracking permission<br>2. Use apps for 2 hours<br>3. Check dashboard | 1.Yes<br>2.Facebook, YouTube | 1. Permission accepted<br>2. Screen time recorded<br>3. Usage breakdown shown | As expected | Pass |

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|---|---|
| **Project Name:** BreathEase | **Test Designed by: Rahat** |
| **Test Case ID:** TC_10 | Test Designed date: 12/01/2026 |
| **Test Priority (Low, Medium, High): High** | Test Executed by: Sheikh Chandni |
| **Module Name:** Doctor's Consultant | Test Execution date: 15/01/2026 |
| **Test Title:** Verify secure consultation | |
| **Description:** Ensure users can securely chat/call with therapist | |
| **Precondition:** Internet connection required | |
| **Dependencies:** Therapist available online. | |

| Test Steps | Test Data | Expected Results | Actual Results | Status (Pass/Fail) |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1. Navigate to Consultant page<br>2. Start chat with therapist<br>3. Call therapist<br>4.working time | 1. Message: “Hello”<br>2. Video call | 1. Page loads<br>2. Chat established<br>3. Call connected securely | As expected | Pass |

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|---|---|
| **Project Name:** BreathEase | **Test Designed by: Sheikh Chandni** |
| **Test Case ID:** TC_11 | Test Designed date: 12/01/2026 |
| **Test Priority (Low, Medium, High): High** | Test Executed by: Prantik paul |
| **Module Name:** Appointment | Test Execution date: 15/01/2026 |
| **Test Title:** Verify Appointment Booking Functionality | |
| **Description:** Verify that a user can pick time slot | |
| **Precondition:** User must be logged in. | |
| **Dependencies:** Appointment slots must be available. | |

| **Test Steps** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |
|---|---|---|---|---|
| 1. Open Appointment booking page<br>2. Select date, and time<br>3. Click Book Appointmen | 1.Date: 22/01/2026 Time: 10:00 AM<br>2. Valid data | 1. Appointment booking page is displayed<br>2. Selected slot is available | As expected | Pass |

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|---|---|
| **Project Name:** BreathEase | **Test Designed by: Jahed Rakib** |
| **Test Case ID:** TC_12 | Test Designed date: 12/01/2026 |
| **Test Priority (Low, Medium, High): High** | Test Executed by: Prantik paul |

| | |
|---|---|
| **Module Name:** Appointment | Test Execution date: 15/01/2026 |
| **Test Title: Appointment Success Message** | |
| **Description:** Verify that a success message is displayed after an appointment is booked successfully**.** | |
| **Precondition:** User must be logged in. | |
| **Dependencies:** Available slot | |

| **Test Steps** | **Test Data** | **Expected Results** | **Actual Results** | **Status (Pass/Fail)** |
|---|---|---|---|---|
| 1.Select service, date, and time<br>2.Click “Confirm / Book Appointment” | 1.Valid appointment details<br>2. Valid data | 1. Appointment booking completed successfully<br>2.Success message displayed: **“**Appointment booked successfully” | As expected | Pass |

## 8. CONCLUSION

The BreathEase project shows how technology can be used positively to improve mental health and digital habits. By tracking user behavior, giving smart alerts, and offering tools like yoga, breathing exercises, mood tracking, and anonymous consultation, the app works as a complete digital wellness assistant. We used the Scrum model to build the system step by step, tested all main features, and made sure they work as expected. The results prove that BreathEase is useful, user-friendly, and effective in reducing stress, anxiety, and digital addiction.